

Flavonoids derived from herbal *Epimedium Brevicornum Maxim* prevent OVX-induced osteoporosis in rats independent of its enhancement in intestinal calcium absorption

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Abstract

Aim: Factorial design was used to test our hypothesis whether a group of flavonoids (FE) derived from herbal *Epimedium Brevicornum Maxim* exerted its preventive effects on estrogen-deficiency-induced osteoporosis mainly through an enhancement in intestinal calcium absorption.

Materials and methods: Forty-five 12-month-old female Wistar rats were used and randomly assigned into sham-operated group and four ovariectomy (OVX) subgroups, i.e. OVX with vehicle (OVX group), OVX with FE (FE group), OVX with calcium supplement (CS group), and OVX with FE and CS (FE + CS group). Daily oral administration of FE (10 mg/kg/day) and/or CS (56 mg/kg/day) started on day 4 after OVX for 12 weeks. Before sacrificing the animals, urine and serum samples were collected for assaying indicators related to intestinal calcium absorption, regulator for calcium homeostasis, and markers of bone turnover. The left proximal femur was dissected for evaluation of the primary end-point (failure force), the second end-points (pQCT-quantified densitometry, geometry, and micro-CT-quantified 3-D trabecula micro-architecture), and pQCT-defined cross-sectional envelope.

Results: FE was found to be able to prevent OVX-induced reduction in failure force as well as the above second end-points, without resulting in an increased uterus weight. CS had no preventive effect on OVX-induced reduction in failure force. Two-way factorial interaction analysis between FE and CS showed that the un-enhanced suppression of parathyroid hormone for calcium homeostasis did not provide link between the enhanced intestinal calcium absorption and the enhanced inhibition of bone resorption in the present study. Furthermore, the discrepancies between the enhanced intestinal calcium absorption and the un-enhanced end-point measures as well as anabolic effect were also revealed by the interaction analysis.

Conclusion: The present study suggested that FE inhibited bone resorption, stimulated bone formation, and accordingly prevented osteoporosis without hyperplastic effect on uterus in the OVX rat model, which was however independent of an enhancement in intestinal calcium absorption.

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Keywords: Herb-derived flavonoids; Intestinal calcium absorption; pQCT and micro-CT; Mechanical test; Bone turnover; Calcium homeostasis

Introduction

It is well known that calcium supplement (CS, a fundamental therapy) and hormone replacement therapy (HRT, a classical therapy) are effective to attenuate bone loss and reduce osteoporotic fracture risk in postmenopausal women [21,31].

A meta-analysis of clinical trials demonstrated that the increase in bone mineral density (BMD) was found 2.5- and 2.7-folds higher at the lumbar spine and femoral neck, respectively, when HRT was administrated together with CS as compared to HRT alone. This suggested that HRT exerted its anti-osteoporosis effect through at least a certain mechanism associated with an enhanced intestinal calcium absorption in postmenopausal women [12,14,21]. However, long-term acceptance and/or compliance of HRT is low due to potentially malignant effects

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on reproductive tissues, which has been recently revealed by the findings from the Women's Health Initiative (WHI) Trial [28,31]. Consequently, there is a need to develop safe therapies alternatively to HRT for prevention of osteoporosis in postmenopausal women [10,29].

Epimedium Brevicornum Maxim has been traditionally used as a medicinal herb to treat fractures, bone and joint diseases, and gonad dysfunctions in Asia for thousands of years [22]. Now, a group of flavonoids (FE) derived from *Epimedium Brevicornum Maxim*, which mainly consists of three phytoestrogenic compounds (Icariin, Genistein, and Daidzein) (Figs. 1-1, 1-2) by a series of standardized extraction and isolation procedures [17,19,22], is developed successfully as an alternative preparation to relieve menopausal symptoms and prevent postmenopausal osteoporosis without hyperplastic effect on the reproductive tissues recently reported in our clinical and experimental studies [22]. However, the mechanism of action of FE has not yet been well investigated. It is known that phytoestrogenic compounds have structural similarities to estrogen conformation and binding capabilities to estrogen receptors, which may therefore promote calcium absorption through an estrogen receptor pathway within intestinal cells [1,2,25] or may regulate bone remodeling via estrogen receptor pathway within bone cells [7,24]. It was however largely unknown whether FE exerting its preventive effect on development of estrogen-deficiency-induced osteoporosis was dependent on an enhanced intestinal calcium absorption or more dependent on a direct action on bone cells.

The present study was therefore performed using a 2×2 factorial design to examine the effect of separate and combined therapy of FE and CS on indicators of intestinal calcium absorption, regulator linked to bone turnover for calcium homeostasis, markers of bone turnover, and end-point measures using ovariectomized (OVX) rat model [22,23]. Serum calcium (S-Ca) and urinary calcium (U-Ca) were used as indicators of intestinal calcium absorption, and serum parathyroid hormone (PTH) was used as regulator for calcium homeostasis, and serum osteocalcin (OC) and urinary deoxypyridinoline (DPD)

were used as bone turnover markers. The primary end-point measure (failure force) and the second end-point measures (densitometry, geometry, and micro-architecture) were evaluated by using state-of-the-art research tools, including mechanical testing in fall configuration, multi-slice peripheral quantitative computed tomography (pQCT), and high resolution micro-computed tomography (micro-CT) [23].

The hypothesis of this study was that, if the skeletal effects of FE were mainly mediated through an enhancement in intestinal calcium absorption, one would expect that interaction between FE and CS resulted in an enhanced indicator of intestinal calcium absorption and subsequently an enhanced suppression of calcium homeostasis regulator PTH linked to an enhanced inhibition of bone resorption and eventually an enhanced primary mechanical end-point at least.

Materials and methods

Animals, groups, treatment, and sampling

Forty-five 12-month-old female Wistar rats (SIPPR-BK Experimental Animal Ltd., China) (body weight 440.1 ± 16.0 g) were housed in animal house at 22°C and with a 12-h light and 12-h dark cycle. During the experimental period, all the rats were fed with standard rat chow (SIPPR-BK Experimental Animal Ltd., China) that contained 0.7% phosphorus, 0.9% calcium, and water ad libitum. Animals were body-weight-matched and randomly assigned into the following five groups: (1) SHAM-operated rats with vehicle (SHAM group), (2) OVX rats with vehicle (OVX group), (3) OVX rats with FE (FE group, 10 mg FE/kg body weight/day), (4) OVX rats with both CS and vehicle (CS group, 56 mg calcium/kg body weight/day), and (5) OVX rats with a combination of FE and CS (FE + CS group, 10 mg FE/kg body weight/day and 56 mg calcium/kg body weight/day). According to the Human–Rat Equivalent Dose Conversion Principle [11,15], the experimental dose for FE and CS in the present study was equivalent to the corresponding clinical prescription dose for a 60-kg human subject. Vehicle, FE, and CS were all administered orally through a custom-made stomach tube, which started on the day 4 after OVX for 12 weeks.

FE (10 mg flavonoids preparation containing 5000 µg Icariin, 250 µg Genistein, 1250 µg Daidzein, and 3500 µg vehicle) was prepared by Tong Ji Tang Pharmacal Company (Gui Zhou, China) [22,23]. The supplemental calcium in form of citrate was from Mission Pharmacal Company (TX, USA).

The body weight of the animals was recorded weekly for adjusting administration of the FE and CS. The body weight before sacrifice was also documented. Urine sample was collected from the rat that was housed individually for 24 h in metabolic cages without providing food and water 1 day before euthanizing the animals. After laparotomy using Ketamine and Xylazine (intramuscularly, 100 mg/kg body weight and 5 mg/kg body weight, respectively), blood sample was collected via abdominal aorta puncture for serum isolation. Urine and serum samples were then stored at -80°C for assaying the biomarkers of intestinal calcium absorption, calcium homeostasis, and bone turnover. Commercially available kits for S-Ca, U-Ca, PTH, OC, and DPD were used.

Animals were euthanized by overdose of Ketamine at the end of 12-week treatment. Uterine was isolated and weighed. The left proximal femur was dissected for measurement of volumetric BMD, geometry and envelope by pQCT, trabecular micro-architecture by micro-CT, and failure force by mechanical testing. Animal ethic approval was obtained from the Chinese University of Hong Kong (CUHK4097/01 M).

Evaluation of indicators of intestinal calcium absorption

Serum calcium (S-Ca) and urine calcium (U-Ca) were analyzed by atomic absorption spectrophotometry (Perkin Elmer Model 503, Norwalk, CT, USA) after diluting aliquots with lanthanum solution [32]. Urinary creatinine (Cr) was determined by a modified Jaffé's method in which picric acid formed a colored

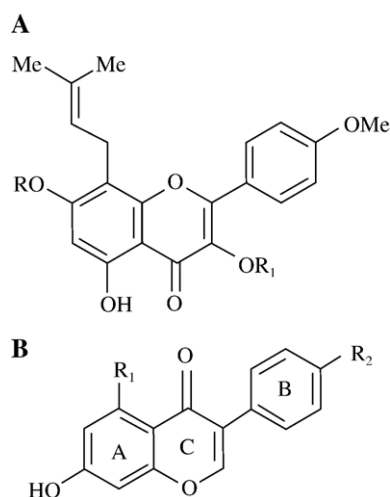


Fig. 1. (A) Structure of Icariin: R = Glucosyl, R₁ = Rhamnosyl. (B) Structure of Daidzein (left) and Genistein (right).

solution in the presence of Cr [8]. Urinary excretion of calcium was expressed as the ratio to Cr excretion.

Evaluation of regulator for calcium homeostasis

Serum parathyroid hormone (PTH) concentration was determined by the immuno-radiometric assay kit which specifically detect the N-terminal and intact PTH (Immutopics, San Clemente, CA, USA). According to the manufacturer, the intra- and interassay precision error was 4.5% and 8.3%, respectively.

Evaluation of bone turnover markers

Serum osteocalcin (OC) concentration was determined by radio-immunoassay kit (Biochemical Technologies, Stoughton, MA, USA), with a 6.8% intra- and 8.9% interassay precision error according to the manufacturer. Urinary deoxypyridinoline (DPD) concentration was determined using HPLC after acid hydrolysis of the urine samples [23]. The intra- and interassay precision error was 3.6% and 8.5%, respectively. Urinary excretion of DPD was expressed as a ratio to Cr excretion.

End-point evaluations for material–geometry–architecture–strength

Mechanical testing

The left femur was tested in a loading configuration designed to simulate a fall onto the greater trochanter [16,36]. Briefly, the distal end of the femur was clamped ante- or retroversion into a custom-made attachment system. The femur was fixed against rotation around the diaphysal axis. An angle of 10° between the femoral shaft and the horizontal platform and an angle of 15° internal rotation in the femoral neck were adjusted for fixation during compressive test. In order to prevent specimen damage, a silicone tube was used to protect the distal end of the specimen and a sheet of rubber was used and placed underneath the trochanter for protection. Compressive test was performed at a speed of 2 mm/min using a material test machine with a 2.5KN load cell (H25KS Hounsfield Test Equipment Ltd., UK). Failure force was recorded as the primary end-point for statistical analysis.

Peripheral quantitative computed tomography (pQCT)

The left proximal femur was fixed with a custom-made plastic holder for rat femur and then scanned at a resolution 220 μm (pixel size) for the base of the femoral neck perpendicularly to the axis of the femoral neck using a multilayer pQCT (Densiscan 2000, Scanco Medical, Bassersdorf, Switzerland). The thickness of each layer was 1 mm. This region was chosen where the fracture occurred during mechanical testing in a fall configuration as reported in our recent studies [22,23,36]. The contours of the periosteum and endosteum were both identified using the built-in software program for measuring densitometry parameters including volumetric BMD of the integral region inside periosteum (iBMD, g/cm³) and volumetric BMD of trabecular region inside endosteum (tBMD, g/cm³). The precision error interpreted in coefficient variation (CV) was reported 1.96% for iBMD and 2.25% for tBMD, respectively [35]. Cross-sectional moment of inertia (CSMI, mm⁴) of cortical bone circumfused by both periosteum and endosteum was also calculated to quantify bone geometry. Both densitometry and geometry were as the second end-point measures. Cross-sectional area circumfused with periosteum (CSPA, mm²) and cross-sectional area circumfused with endosteum (CSEA, mm²) were calculated to quantify corresponding bone envelope, respectively.

Micro-CT

After pQCT measurement, one representative left femur from each group was selected for scanning with micro-CT (μCT-40, Scanco Medical, Bassersdorf, Switzerland). The selection of representative sample was based on the median value of tBMD of respective group [22–23]. Briefly, the femoral neck was aligned perpendicularly to the scanning axis for a total scanning length of 7.9 mm. The scanning was conducted at a resolution of 12 μm per voxel, and the segmentation parameters for bone from background were fixed at Sigma = 1.2, Support = 2, and Threshold = 214. The trabecular bone within the greater trochanter was extracted with semi-automatically drawn contour at each two-dimensional (2-D) section. The volume of interest inside the greater trochanter region was located where femoral neck just met greater trochanter and

the 80 continuous slices above were chosen for data acquisition. The three-dimensional reconstructed images were used directly to quantify micro-architecture as the second end-point measure with the image analysis program of the micro-CT workstation [23,28].

Statistical analysis

The results were presented as mean ± standard deviation. LSD post hoc test was made for multiple comparisons among all five groups if one-way ANOVA test was found to be statistically significant. The data for all four OVX subgroups were analyzed using a two-factorial ANOVA with body weight as a covariant variable for eliminating its influence on the measurement results, where the two treatment factors were FE (presence or absence) and CS (presence or absence). In this way, it was determined whether the two different treatments mutually influenced each other in an interaction way (i.e. not a simply additive way). There are two different interaction ways. One is synergetic, the other is antagonistic. A synergetic interaction between FE and CS was expressed using the following formula (1), and the antagonistic interaction was expressed using the following formula (2).

$$‘FE + CS’ - ‘OVX’ > (‘FE’ - ‘OVX’) + (‘CS’ - ‘OVX’) \quad (1)$$

$$‘FE + CS’ - ‘OVX’ < (‘FE’ - ‘OVX’) + (‘CS’ - ‘OVX’) \quad (2)$$

Statistical significance was set at $P < 0.05$. All statistical analysis was performed using SPSS 11.0 (SPSS Inc, Chicago, IL, USA).

Results

Body weight and uterine weight

The OVX group showed 15% significantly higher body weight but 80% lower uterine weight than the SHAM group ($P < 0.05$ for both). Body weight and uterine weight were influenced neither by FE nor by CS when compared to the OVX group ($P > 0.05$ for both). No interaction was found between FE and CS on both body weight and uterine weight ($P > 0.05$ for both). Multiple comparison showed no difference in both body weight and uterine weight among FE, CS, and FE + CS group ($P > 0.05$) (Table 1).

Intestinal calcium absorption and calcium homeostasis

The measurement values of PTH, S-Ca, and U-Ca did not show significant differences between the SHAM and OVX group ($P > 0.05$ for all). Compared to the OVX group, PTH was decreased by 36% in FE group and by 35% in CS group ($P < 0.05$ for both). The decrease in PTH was 39% in FE + CS group, which did not demonstrate any interaction between FE and CS treatment ($P > 0.05$). Both FE and CS treatment showed 56% and 44% increase in U-Ca, respectively ($P < 0.05$ for both). The combined treatment of FE and CS (FE + CS group) resulted in 110% increase in U-Ca, which revealed a synergetic interaction between FE and CS by interaction analysis ($P < 0.05$). For the S-Ca level, the FE group and CS group showed on average 18% and 16% increase, respectively ($P < 0.05$ for both). However, the FE + CS group only resulted in 17% increase in S-Ca, without interaction between FE and CS treatment ($P > 0.05$). Multiple comparisons among the FE, CS, and FE + CS group showed no difference in both S-Ca and PTH ($P > 0.05$). However, there was a significantly higher U-Ca

Table 1
Summary of results for body weight, uterine weight, pQCT scan, mechanical testing, and biochemical assay

Parameters			SHAM	OVX	FE	CS	FE + CS	Two-way factorial ANOVA (<i>P</i> value)			One-way ANOVA-based multiple comparisons (<i>P</i> value)		
								Main effects		Interaction effects			
								FE vs. OVX	CS vs. OVX	FE x CS	FE vs. CS	FE vs. FE + CS	CS vs. FE + CS
Body weight (g)			497 ± 19.53	572 ± 22.49**	564 ± 20.34	573 ± 22.04	560 ± 22.46	<i>P</i> = 0.22	<i>P</i> = 0.84	<i>P</i> = 0.79	<i>P</i> > 0.05		
Uterine weight (mg)			1100 ± 43.20	220 ± 8.65**	222 ± 8.12	226 ± 8.70	228 ± 9.16	<i>P</i> = 0.46	<i>P</i> = 0.18	<i>P</i> = 0.99	<i>P</i> > 0.05		
pQCT Scan	Densitometry	iBMD (g/cm ³)	1.21 ± 0.07	0.91 ± 0.08**	1.20 ± 0.04	1.10 ± 0.08	1.21 ± 0.08	<i>P</i> = 0.00	<i>P</i> = 0.00	<i>P</i> = 0.10	<i>P</i> > 0.05		
		tBMD (g/cm ³)	1.20 ± 0.07	0.89 ± 0.07**	1.19 ± 0.12	1.09 ± 0.09	1.20 ± 0.068	<i>P</i> = 0.00	<i>P</i> = 0.00	<i>P</i> = 0.11	<i>P</i> > 0.05		
	Geometry	CSMI (mm ⁴)	21.54 ± 2.38	14.43 ± 3.97**	19.29 ± 1.50	18.57 ± 1.21	20.71 ± 2.12	<i>P</i> = 0.00	<i>P</i> = 0.00	<i>P</i> = 0.36	<i>P</i> < 0.05	<i>P</i> > 0.05	<i>P</i> < 0.05
		Envelope	CSPA (mm ²)	15.90 ± 1.30	16.05 ± 1.19	17.33 ± 0.88	15.89 ± 1.17	17.16 ± 0.87	<i>P</i> = 0.00	<i>P</i> = 0.67	<i>P</i> = 0.99	<i>P</i> < 0.05	<i>P</i> > 0.05
		CSEA (mm ²)	2.43 ± 2.38	4.90 ± 0.10**	4.15 ± 0.69	4.44 ± 0.10	2.92 ± 0.54	<i>P</i> = 0.00	<i>P</i> = 0.00	<i>P</i> = 0.03 ¹	<i>P</i> > 0.05	<i>P</i> < 0.05	<i>P</i> < 0.05
		Mechanical Testing property	Failure force (N)	83.57 ± 11.26	69.56 ± 10.74*	84.89 ± 8.04	78.00 ± 15.54	86.16 ± 20.27	<i>P</i> = 0.00	<i>P</i> = 0.27	<i>P</i> = 0.14	<i>P</i> > 0.05	
Biochemical Assay	Intestinal calcium absorption	U-Ca/Cr (mmol/mmol)	0.22 ± 0.03	0.18 ± 0.02	0.28 ± 0.05	0.26 ± 0.04	0.38 ± 0.08	<i>P</i> = 0.00	<i>P</i> = 0.00	<i>P</i> = 0.00 ¹	<i>P</i> > 0.05	<i>P</i> < 0.05	<i>P</i> < 0.05
		S-Ca (mmol/l)	2.10 ± 0.59	1.97 ± 0.35	2.32 ± 0.22	2.29 ± 0.21	2.31 ± 0.23	<i>P</i> = 0.03	<i>P</i> = 0.04	<i>P</i> = 0.45	<i>P</i> > 0.05		
	Calcium homeostasis	PTH (pg/ml)	20.90 ± 8.39	26.98 ± 4.45	17.31 ± 6.95	17.39 ± 6.98	16.37 ± 6.57	<i>P</i> = 0.03	<i>P</i> = 0.04	<i>P</i> = 0.20	<i>P</i> > 0.05		
		Bone turnover	OC (nmol/l)	5.47 ± 0.21	6.29 ± 0.25**	6.20 ± 0.23	5.80 ± 0.20	6.09 ± 0.21	<i>P</i> = 0.22	<i>P</i> = 0.00	<i>P</i> = 0.03 ²	<i>P</i> < 0.05	<i>P</i> > 0.05
		DPD/Cr (nmol/mmol)	63.20 ± 25.37	85.94 ± 9.31**	67.24 ± 11.09	78.06 ± 1.81	47.79 ± 8.09	<i>P</i> = 0.00	<i>P</i> = 0.00	<i>P</i> = 0.02 ¹	<i>P</i> > 0.05	<i>P</i> < 0.05	<i>P</i> < 0.05

Note. **P* < 0.05, ***P* < 0.01 vs. OVX. Statistically multiple comparisons among all five groups were made using one-way ANOVA with post hoc test. The prevention effect of FE and CS and their interaction effect in the four OVX subgroups were analyzed by two-way factorial ANOVA test. 1: synergetic interaction. 2: antagonistic interaction.

Table 2
Micro-CT 3-D parameters of trabecula bone in greater trochanter region

Group	BV/TV (1)	Conn.D (1/mm ³)	SMI (1)	Tb.N (1/mm)	Tb.Th (mm)	Tb.Sp (mm)
SHAM	0.24	18.43	0.26	2.26	0.09	0.44
OVX	0.12	15.42	1.05	1.92	0.08	0.52
FE	0.20	22.11	0.77	2.19	0.11*	0.47
CS	0.16	16.87	1.01	1.94	0.09	0.52
FE + CS	0.19	22.75	0.96	2.21	0.10	0.43

Note. * Tb.Th in the FE rat, exceeding that in the SHAM rat, was found highest among all the representative samples.

level in the FE + CS group than that in the FE group or CS group ($P < 0.05$), whereas no difference was found between the FE group and CS group ($P > 0.05$) (Table 1).

Bone turnover

As compared with SHAM group, the OVX group showed on average 15% higher OC level and 36% higher DPD level ($P < 0.05$ for both). When comparison was made with OVX group, DPD level decreased 22% and 9% in FE group and CS group, respectively ($P < 0.05$ for both). The FE + CS treatment resulted in a further decrease of 44% in DPD, in which a synergetic interaction between FE and CS was found by interaction analysis ($P < 0.05$). In analysis of OC, the CS group showed 8% decrease ($P < 0.05$), whereas the FE group showed no change ($P > 0.05$). The combined treatment with FE + CS showed a decrease of 3%, in which an antagonistic interaction effect between FE and CS was found ($P < 0.05$). Multiple comparisons showed a significantly lower OC level in the CS group than that in the FE group or FE + CS group ($P < 0.05$), whereas no difference was found between the FE group and FE + CS group ($P > 0.05$). For DPD, a significantly lower level was found in the FE + CS group than that in the FE group or CS group ($P < 0.05$), whereas no difference was found between the FE group and CS group ($P > 0.05$) (Table 1).

Mechanical testing

The OVX group showed on average 17% lower failure force than that of the SHAM group ($P < 0.05$). Compared to the OVX group, failure force was found 22% higher in the FE group ($P < 0.05$), while 12% higher failure force in CS group was not found to be statistically significant ($P > 0.05$). The FE + CS

group showed 24% higher failure force, which was however not evidenced with an interaction between FE and CS ($P > 0.05$). Multiple comparisons made among the FE, CS, and FE + CS group showed no difference in failure force ($P > 0.05$) (Table 1).

pQCT evaluation

Densitometry and geometry

The OVX group showed significantly lower volumetric BMD (25% lower iBMD and 26% lower tBMD) and 33% less CSMI than that of the SHAM group ($P < 0.05$ for all). However, iBMD, tBMD, and CSMI were found significantly higher in the FE and CS group than those in the OVX group ($P < 0.05$ for all), yet without interaction observed between FE and CS treatment ($P > 0.05$). Multiple comparisons made among the FE, CS, and FE + CS group showed no difference in above variables ($P > 0.05$) (Table 1).

Envelope

The OVX group showed on average 101% higher CSEA than the SHAM group ($P < 0.05$). CSPA of the OVX group was found not significantly but slightly higher as compared with that of the SHAM group ($P > 0.05$). As compared to the OVX group, CSPA was found significantly higher in the FE group ($P < 0.05$) but not in the CS group ($P > 0.05$), and no interaction was found between FE and CS treatment ($P > 0.05$). CSEA was shown 15% and 9% decrease in FE group and CS group, respectively ($P < 0.05$ for both). The FE + CS group resulted in a further decrease of 40%, and interaction analysis demonstrated their synergetic interaction effect on CSEA between FE and CS ($P < 0.05$). Multiple comparisons showed a significantly lower CSPA in the CS group than that in the FE group or FE + CS group ($P < 0.05$), whereas no difference was found between the FE group and FE + CS group ($P > 0.05$). A significantly lower CSEA was found in the FE + CS group than that in the FE group or CS group ($P < 0.05$), whereas no difference was found between the FE group and CS group ($P > 0.05$) (Table 1).

Micro-CT evaluation

The representative samples showed that the vehicle-treated OVX rat had lower value in bone volume/tissue volume (BV/TV), connect density (Conn.D), trabecula number (Tb.N),

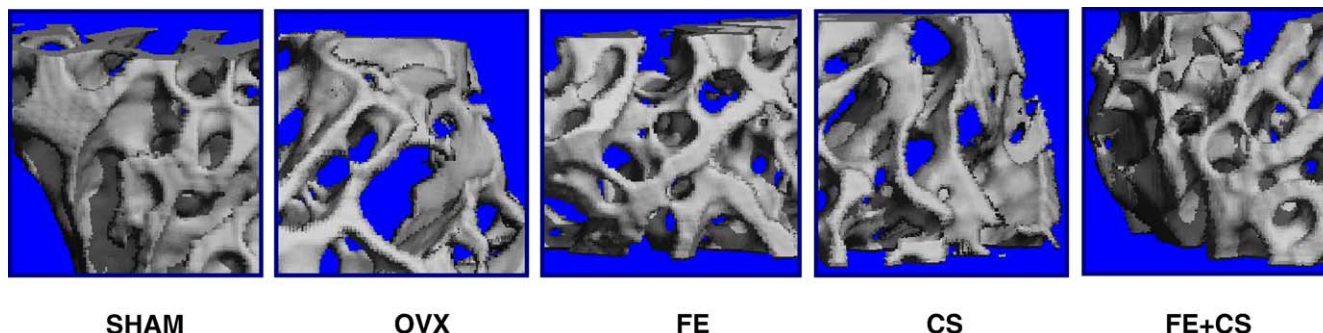


Fig. 2. Representative sample from each group: 3-D architecture of trabecula bone within greater trochanter region.

trabecula thickness (Tb.Th), and higher value in structure model index (SMI) and trabecula space (Tb.Sp), when compared to the vehicle-treated SHAM rat (Table 2, Fig. 2).

Compared to the vehicle-treated OVX rat, the FE treatment and FE + CS treatment reversed the abovementioned findings at a same degree. Tb.Th in the FE rat was the highest among all the representative samples, which even exceeded that of the SHAM rat (Table 2).

Discussion

Using the state-of-the-art techniques and factorial design statistics, this study revealed for the first time that the potential beneficial effects of flavonoids (FE) from herbal *Epimedium Brevicornum Maxim* on prevention of OVX-induced osteoporosis were independent of an enhancement in intestinal calcium absorption.

Un-enhanced suppression of PTH for calcium homeostasis did not provide intermedium link between the enhanced intestinal calcium absorption and the enhanced inhibition of bone resorption in the present study

The results showed a larger CSEA accompanied by a higher DPD level in the OVX group when compared to the SHAM group, suggesting OVX-induced endosteum bone resorption [9]. In contrast, FE demonstrated its inhibition effect on endosteum bone resorption, as evidenced by a smaller CSEA accompanied by a lower DPD level in the FE group when compared to the OVX group. Furthermore, synergetic interaction effects on both CSEA and DPD level were found between FE and CS, that is, the inhibition effect of FE on OVX-induced bone resorption was enhanced by supplemental calcium. On the other hand, enhanced intestinal calcium absorption by FE was documented by a higher U-Ca excretion level in the FE group when compared to the OVX group and a synergetic interaction effect on stimulation of U-Ca excretion level between FE and CS.

It has been known that regulation for calcium homeostasis commonly gives a medium link between intestinal calcium absorption and bone turnover [20]. In the present study, PTH for calcium homeostasis, which is one of the most important stimulators of osteoclastic bone resorption [5,20], was found lower in the FE group when compared to the OVX group. If the inhibition effect of FE on bone resorption was mainly mediated through enhancing intestinal calcium absorption, one would expect a synergetic interaction effect on stimulation of U-Ca excretion between FE and CS, and subsequently their synergetic interaction effect on suppression of PTH, and then their synergetic interaction effect on inhibition of bone resorption. However, no significant interaction was found on PTH between FE and CS. Thus, a causal relation between enhanced inhibition of OVX-induced bone resorption and enhanced intestinal calcium absorption could not be established for FE due to un-enhanced suppression of PTH, that is, the inhibition effect of FE on bone resorption could not be dependently explained by its enhancement in intestinal calcium absorption. It was unclear at

present how FE inhibited OVX-induced bone resorption. Nevertheless, it strongly indicated that an enhancement in intestinal calcium absorption was not the main pathway for FE inhibiting OVX-induced bone resorption.

Discrepancy between the enhanced intestinal calcium absorption and un-enhanced anabolic effect on bone envelop was found in the present study

OVX induced a slightly higher value in CSTA when compared to the SHAM rats, which might suggest that OVX-induced moderate modeling-dependent bone formation at periosteum aimed at maintaining bone strength for compensation of bone loss [3,6,9]. Furthermore, FE showed its anabolic effect on periosteum during estrogen deficiency as evidenced by a larger CSPA measured by pQCT, which was accompanied by unreduced OC in the FE group when compared to the OVX group. On the other hand, favorable three-dimensional architecture of trabecular bone evaluated by micro-CT was found in the FE sample as compared to the vehicle-treated OVX sample. The Tb.Th in the FE rat was even 22% higher than that in the SHAM rat, which was consistent with the findings of our recent study [22]. As 15-month-old female Wistar rats already reached mature statues [4,22], the increase in the Tb.Th in response to FE observed in the present study cannot be explained by its anti-resorptive activity through filling remodeling space [30]. Rather, this effect may be only viewed as bone resorption exceeded by bone formation, i.e. the anabolic effect of FE on trabecular bone.

Inferentially, if the anabolic effect of FE on periosteum and trabecula was mainly explained by the above-documented enhancement in intestinal calcium absorption, one would expect an enhanced effect on stimulating modeling-dependent bone formation at periosteum and trabecula. However, no significant interaction effect on CSPA as well as Tb.Th was found between FE and CS. This implied that the anabolic effect of FE on periosteum and trabecula could not be enhanced by supplemental calcium and hence suggested that FE exerted its anabolic effect not mainly through an enhancement in intestinal calcium absorption.

Discrepancy between enhanced intestinal calcium absorption and un-enhanced end-point bone measures was found in the present study

If FE exerted its prevention effect on OVX-induced osteoporosis through an enhancement in intestinal calcium absorption, one would accordingly expect a significant interaction effect on end-point bone indexes such as the primary end-point measure (failure force) and the second end-point measures (densitometry, geometry, and micro-architecture). However, effective prevention was not found in the CS group but in the FE group, as evaluated by the mechanical testing for failure force (the primary end-point measure), and there was no interaction between FE and CS, that is, supplemental calcium did not enhance the prevention effect of FE on OVX-induced reduction in failure force. Furthermore, no interaction effects on the second

end-points, i.e. the bone material and structural parameters evaluated by pQCT as well as micro-CT, were found between FE and CS. This implied that FE administration exerting its preventive effect on OVX-induced osteoporosis was independent of an enhancement in intestinal calcium absorption.

A direct action by FE on bone cells was proposed

Recently, *in vitro* evidences have showed that FE compounds, such as Icaritin, Genistein, and Daidzein, could stimulate osteogenesis in osteoblast and/or osteoprogenitor cells and inhibit bone resorption via direct targeting of osteoclast progenitors through estrogen receptor pathways [19,24,25]. On the other hand, it has also been reported that estrogen receptor beta pathway can be linked to endosteal resorption and periosteal apposition of new bone, whereas estrogen receptor alpha pathway could be linked to the preservative effects on trabecular surfaces [13,18,26,33]. Due to diversities of FE compounds (Icaritin, Genistein, and Daidzein) in chemical structure, metabolism, and affinity for estrogen receptor [27,34], FE might be involved in a direct action on bone cells through multiple estrogen receptor sub-type pathways.

In conclusion, the present study is contrary to the hypothesis. Through a pathway independent of intestinal calcium absorption, FE inhibited bone resorption at endosteum/trabecula, stimulated bone formation at periosteum/trabecula, and accordingly prevented osteoporosis without hyperplastic effect on uterus in the OVX rat model. More specifically, designed research is required to further elucidate the mechanisms of FE actions for both anti-resorption and anabolic effect.

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